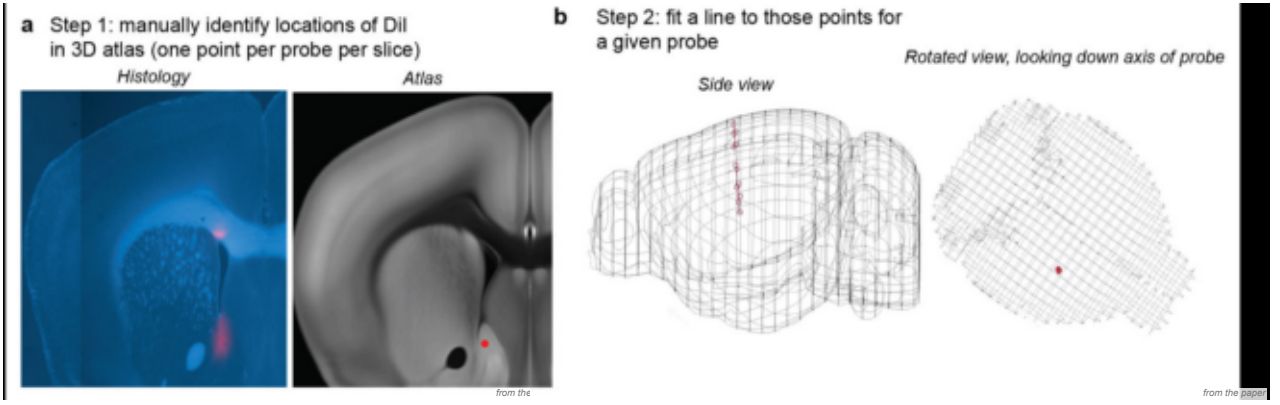
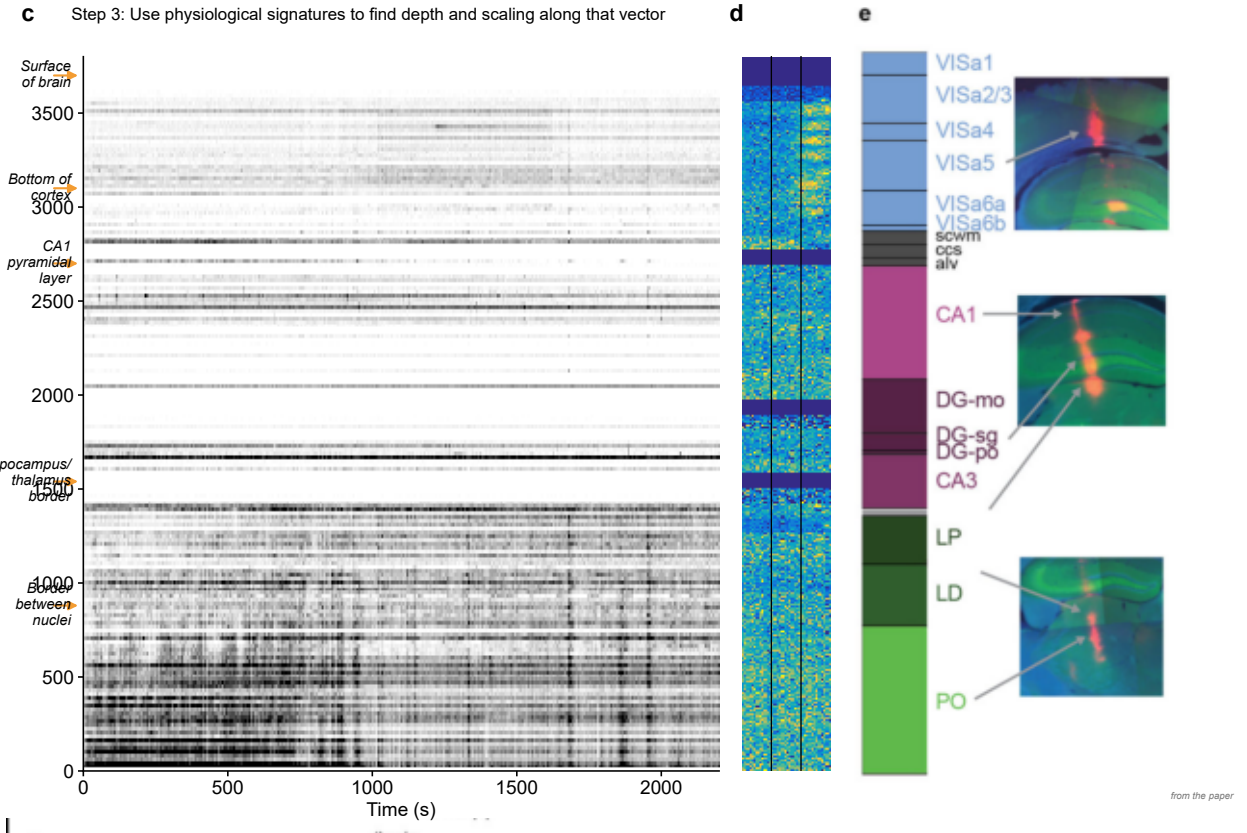
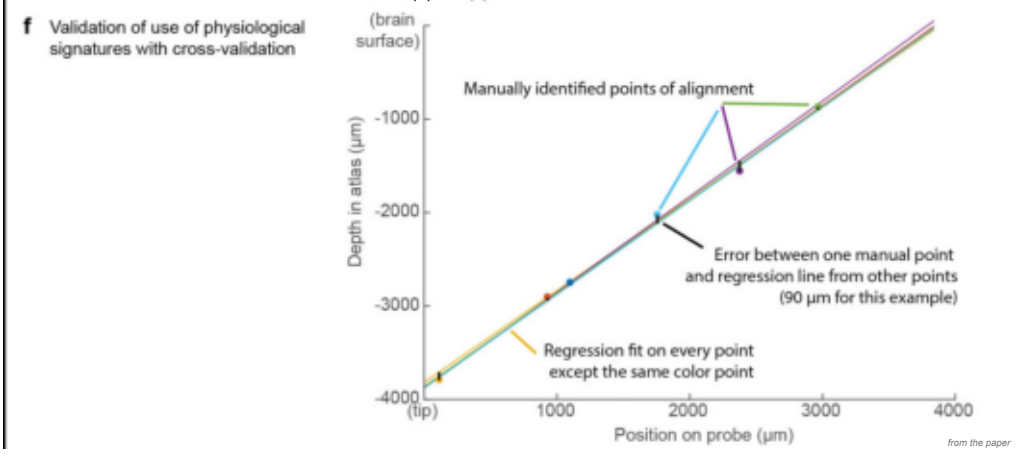

a Step 1: manually identify locations of DiI in 3D atlas (one point per probe per slice)
Histology
Atlas
b Step 2: fit a line to those points for a given probe
Side view
Rotated view, looking down axis of probe
from the paper


c Step 3: Use physiological signatures to find depth and scaling along that vector
Surface of brain
Bottom of cortex
CA1 pyramidal layer
Hippocampus/ thalamus border
Border between nuclei
3500
3000
2500
2000
1500
1000
500
0
0
500
1000
1500
2000
Time (s)
d
e
VISa1
VISa2/3
VISa4
VISa5
VISa6a
VISa6b
scwm
ccs
alv
CA1
DG-mo
DG-sg
DG-po
CA3
LP
LD
PO
from the paper


f Validation of use of physiological signatures with cross-validation
(brain surface)
Manually identified points of alignment
-1000
-2000
-3000
-4000
Depth in atlas (μm)
Error between one manual point and regression line from other points (90 μm for this example)
Regression fit on every point except the same color point
(tip)
1000
2000
3000
4000
Position on probe (μm)
from the paper